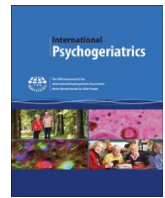




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Review article

Diagnosis, evaluation & management of cognitive disorders in Parkinson's disease: A systematic review

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ABSTRACT

Background: There is considerable variability in the diagnosis, evaluation and management of cognitive disorders in Parkinson's disease (PD) across clinical services. A review of guidelines and relevant literature will provide recommendations to guide clinical decision-making. The present review aimed to summarise and critically appraise current recommendations for the diagnosis, evaluation and management of cognitive disorders in PD.

Method: Five academic databases (PubMed, SCOPUS, Medline, PsycINFO, CINAHL) and five grey literature databases were systematically searched in August 2024 by two independent reviewers following PRISMA guidelines. Guidelines and systematic reviews from 2003–2024 available in English and reporting at least one relevant recommendation for the diagnosis, evaluation or management of cognitive disorders in PD were included. Quality assessment was completed using the AGREE-II tool for guidelines and AMSTAR tool for systematic reviews.

Results: In total, 32 guidelines and 24 systematic reviews were included. Guideline quality ranged from moderate to high and systematic review quality from critically low to high. Articles provided recommendations for cognitive impairment in PD in terms of diagnosis, neuropsychological evaluation, treatment and care. However, recommendations for cognitive tools, care considerations, and non-pharmacological interventions were limited, despite relevant evidence from the systematic review literature.

Conclusion: Significant gaps identified in processes for neuropsychological evaluations, inconsistent recommendations for non-pharmacological interventions, and limited care considerations calls for future iterations of the clinical practice guidelines for cognitive impairment in PD.

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Introduction

Parkinson's disease (PD) is a progressive, neurodegenerative condition with significant cognitive implications. PD is often diagnosed in neurology or geriatric specialised health services settings where the focus of clinical care remains on motor symptoms, while cognitive symptoms are typically considered secondary. However, cognitive impairment is a primary concern for people living with PD, with 83 % of people living with PD developing dementia by late stages of the disease [1,2].

There is heterogeneity in the type (e.g., memory, executive function) and trajectory (e.g., rapid or gradual decline) of cognitive deficits in PD [1,3]. Mild Cognitive Impairment in PD (PD-MCI) is a transitory state between normal cognition and PD dementia (PDD) where functional ability remains intact [4]. In PDD, impairments are severe and functional independence is lost [5,6]. Up to 60 % of people with PD-MCI progress to PDD within five years [1,7,8]. PD-MCI is analogous to mild neurocognitive disorder from the Diagnostic Statistical Manual 5 (DSM-5-TR), while PDD is analogous to major neurocognitive disorder from the DSM-5-TR [9]. Cognitive disorders significantly increase risk of falls, increase care partner burden, promote earlier institutionalisation, and decrease quality of life for people living with PD [1,10].

Despite negative impact, there is considerable variability in the diagnosis, evaluation and management of cognitive disorders in PD across clinical services. The present systematic review aims to consolidate and critically appraise current guidelines and recommendations for the diagnosis, neuropsychological evaluation, and management of cognitive disorders in PD.

Methodology

Eligibility criteria and search protocol

The "PICAR" framework was adopted to frame the eligibility criteria (Table 1) [11]. Systematic reviews or guidelines that made recommendations for the diagnosis, evaluation or management of cognitive disorders in PD were included. The protocol adhered to PRISMA guidelines [12], (PROSPERO registration: CRD42023449106). The following systematic search terms were used: (Parkinson* disease) AND (cognit* OR neuropsych*) AND (guideline* OR recommend*). Full search terms for each database are provided in the [Supplementary Material](#). Results included publications from January 2003 up to September 2024 from PubMed, SCOPUS, Medline, PsycINFO, CINAHL and five grey literature databases: Guidelines International Network, Canadian Medical Association – Infobase: Clinical Practice Guidelines, Medscape UK, National Institute for Health and Care Excellence (NICE), and New Zealand Guidelines Group. Manual searches of reference lists supplemented the search. The protocol for the project has been approved by the Metro North Health HREC (HREC/2023/MNHA/100098) and it conforms to the provisions of the Declaration of Helsinki.

Table 1
PICAR elements and associated eligibility criteria.

PICAR element	Eligibility criteria	
	Inclusion	Exclusion
Population	People living with PD and cognitive impairment associated with PD	Dementia with Lewy Bodies, Alzheimer's disease, or other types of dementia
Interventions	Diagnosis and management of cognitive impairment	Diagnosis and management of PD or other PD-related symptoms
Comparators	Not applicable	Not applicable
Attributes of eligible guidelines	Original, peer-reviewed research articles or guidelines from 2003 – 2023 that are written in English language within grey and academic literature	Conference abstracts, book chapters, articles unavailable in English
Recommendation characteristics	Articles are only eligible for inclusion if they report at least one recommendation of interest related to diagnosis, neuropsychological evaluation, treatment and/or care of people living with PD and cognitive impairment	Recommendations are not made

Note. PD = Parkinson's disease

Study selection, data extraction and quality assessment

The title/abstract screen was completed by two independent reviewers using Covidence [13], with discrepancies in article selection resolved by a third arbitrator. Full-text articles were evaluated for inclusion and the following data was extracted from included studies: publication details, methodology, recommendations, and limitations. A list of excluded studies are reasons for exclusion are provided in the [Supplementary Material](#). Two reviewers independently applied the Appraisal for Guidelines for Research and Evaluation II (AGREE-II) [14] and the Assessment of Multiple Systematic Reviews (AMSTAR) [15] tools to assess quality of guidelines and systematic reviews, respectively (see [Supplementary Material](#) for scoring matrices).

Results

Article selection and quality appraisal

In total, 32 guidelines and 24 systematic reviews were included (Fig. 1). Most guidelines scored above average (> 4 out of 7) on the AGREE II tool, with only 5 out of 32 articles scoring below this [5,16–19], indicating overall moderate to high quality articles. However, most systematic reviews (19/24) were classified as low or critically low quality on the AMSTAR, mostly due to incomplete reporting of excluded studies and risk of bias.

Article characteristics

Recommendations were made for the diagnosis, neuropsychological assessment, treatment, and care of cognitive disorders in PD. Notably, 13 out of 32 (40 %) guidelines and 21 out of 24 (87 %) systematic reviews were published within the past five years. Seven guidelines were specific to cognitive impairment in PD [4,5,16,20–23], 16 guidelines related to diagnosis and management of PD with a section on cognitive impairment [18,19,24–38], and 9 guidelines were for the diagnosis or management of general conditions (e.g., dementia, neurological disease) with a section on PD [17,39–47]. Guidelines were produced from more than 10 countries, with 15 derived from international collaborations.

Summary of recommendations

Diagnosis

Diagnostic criteria. Currently, cognitive disorders in PD can formally be diagnosed using *Movement Disorder Society* (MDS) criteria for PD-MCI or PDD, or using DSM-5-TR criteria for major or minor neurocognitive disorder due to PD. For the diagnosis of PD-MCI or PDD, MDS guidelines highlighted that the diagnosis of PD should be clinically confirmed [4,5]. MDS guidelines for PDD also stated that the onset of cognitive symptoms

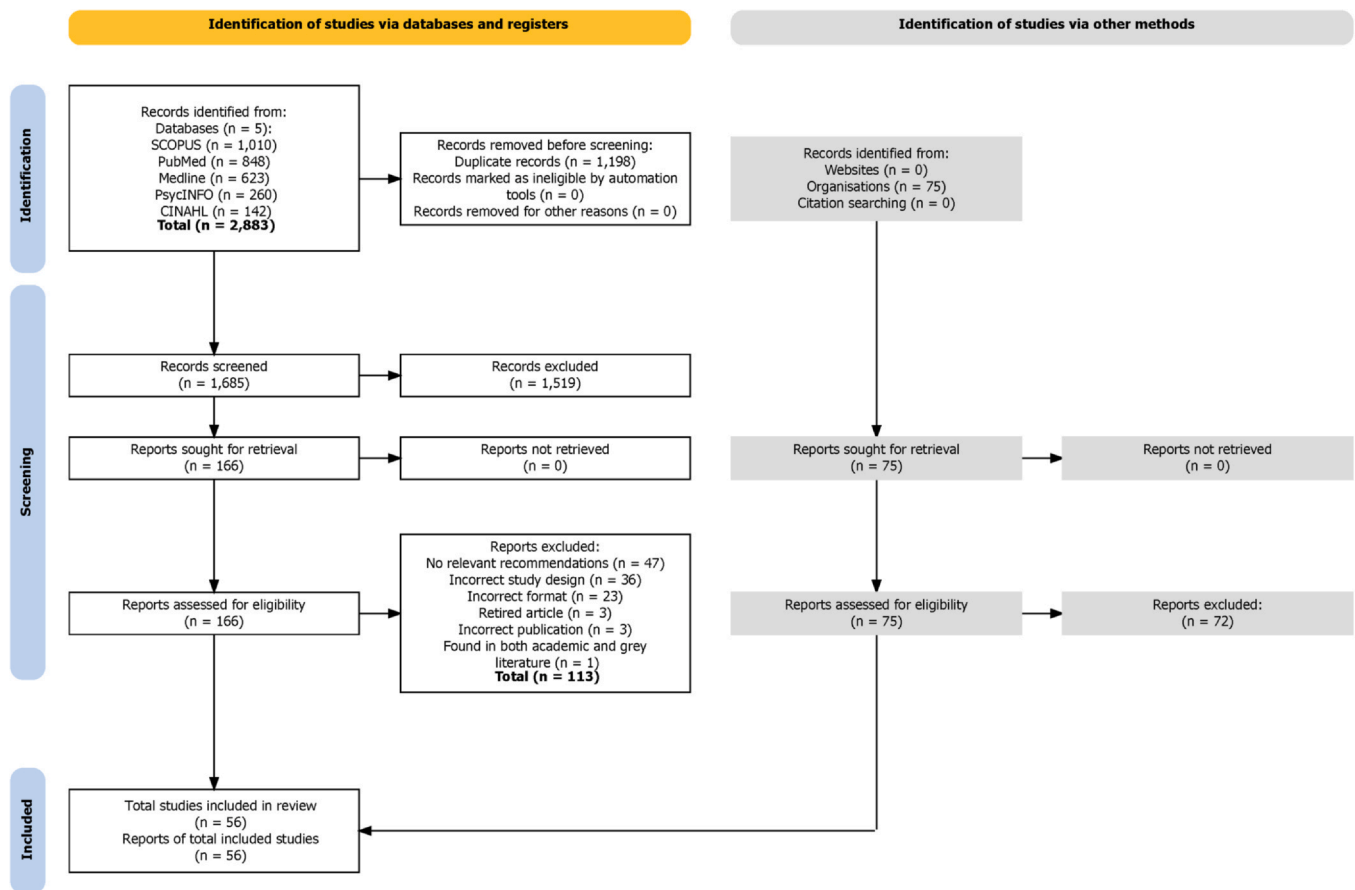


Fig. 1. PRISMA Flow Chart.

appear at least one year after the onset of motor symptoms, as concurrent motor and cognitive symptom onset is indicative of dementia with Lewy Bodies (DLB) [5]. The DSM-5-TR diagnostic criteria for neurocognitive disorder due to PD require that: criteria are met for major or mild neurocognitive disorder; the disturbance occurs in the setting of established PD; there is insidious onset and gradual progression of impairment; and the impairment is not attributable to another medical condition or mental disorder [9]. For comparison, MDS and DSM-5-TR diagnostic criteria are provided in Table 2.

Risk factors. Diagnosis of Subjective Cognitive Decline (SCD) was not endorsed by guidelines, although one systematic review reported associations between SCD and subsequent objective cognitive decline in PD. It was recommended establishing a universal PD-SCD definition of 'self/informant-reported cognitive decline' to identify those at risk of PD-MCI [48]. Similarly, Postuma et al. (2024) recommended risk factors of near-term cognitive impairment, such as age, sleep disorders, and neurogenic orthostatic hypotension, to guide symptomatic treatment decisions in PD [38].

Subtypes. Cognitive subtyping was recommended only by the MDS guidelines for PD-MCI for research purposes [4]. Subtype classifications included single- or multiple-domain PD-MCI with specification of which domains are impaired (e.g., single-domain PD-MCI: language).

Clinical practice. There were some recommendations regarding the process of discussing cognitive impairment in PD in the clinical setting. One guideline highlighted that all people living with PD should be provided with verbal and written communication due to their risk of cognitive impairment [33]. Goldman et al. (2018) recommended providing information at diagnosis regarding the cognitive aspects of

PD and realistic expectations of cognitive decline throughout the disease, as well as discussing advanced care planning [21].

Secondary causes. Two articles recommended administration of routine blood tests to rule out alternative causes of cognitive disorders, such as, vitamin B12, folate, Thyroid Stimulating hormone, liver function, syphilis, Borrelia and HIV [22,44]. Parmera et al. [22] also recommended magnetic resonance imaging (MRI) or computed tomography (CT) scans to rule out vascular dementia, stroke, and brain tumours [22].

Biomarkers. Sorbi et al. (2012) stated that structural or functional MRI, EEG, CSF analysis, genetic testing or biopsy for the diagnosis of dementia in PD is not backed by evidence [44]. Two guidelines recommended that cortical hypometabolism on fludeoxyglucose positron emission tomography (FDG-PET) may be predictive of cognitive decline in PD and contribute to a dementia diagnosis [35,49]. MDS guidelines for PDD did not encourage prescriptive ancillary investigations, instead recommending clinicians to follow up with additional tests only if concerns of potential comorbidities are raised during history taking or clinical examination [5].

Neuropsychological Evaluation

Evaluation criteria. The seminal MDS guidelines for PD-MCI and PDD were most frequently referred to across the literature [4,5]. Eight of the reviewed guidelines addressing neuropsychological evaluation endorsed MDS guidelines [16,20,22,25,26,33,39,44]. Both MDS guidelines recommended two levels of cognitive assessment: level 1 comprising brief cognitive screens, and level 2 comprising comprehensive neuropsychological assessment. PDD criteria stipulate that level 1 be used for a diagnosis of dementia and level 2 be reserved

Table 2

Diagnostic criteria for cognitive disorders in PD.

Citation	Condition	Diagnostic Criteria
Dubois et al. 2007	MDS Task Force Criteria for PDD	1. PD diagnosis 2. Parkinsonism developed before dementia 3. Mini Mental State Exam (MMSE) < 26 4. Dementia has Impact on Activities of Daily Living 5. Impaired cognition (two tests abnormal out of: Months reversed or Sevens backwards, Lexical fluency or clock drawing, MMSE pentagons, MMSE 3-word recall) 6. Absence of Major Depression 7. Absence of delirium 8. Absence of other abnormalities that obscure diagnosis
Litvan et al. 2012	MDS PD-MCI Criteria	Inclusion criteria 1. Diagnosis of PD as based on the UK PD Brain Bank Criteria 2. Gradual decline, in the context of established PD, in cognitive ability reported by either the patient or informant, or observed by the clinician 3. Cognitive deficits on either formal neuropsychological testing or a scale of global cognitive abilities 4. Cognitive deficits are not sufficient to interfere significantly with functional independence, although subtle difficulties on complex functional tasks may be present Exclusion criteria 1. Diagnosis of PD dementia based on MDS Task Force proposed criteria 2. Other primary explanations for cognitive impairment (e.g., delirium, stroke, major depression, metabolic abnormalities, adverse effects of medication, or head trauma) 3. Other PD-associated comorbid conditions (e.g., motor impairment or severe anxiety, depression, excessive daytime sleepiness, or psychosis) that, in the opinion of the clinician, significantly influence cognitive testing Major Neurocognitive disorder (NCD) due to PD is characterised by a significant decline in one or more cognitive domains that interferes with activities of daily living. The motor symptoms must be present prior to the onset of cognitive decline. The cognitive decline should not be due to delirium or other mental disorders. Mild Neurocognitive disorder (NCD) due to PD is characterised by a modest decline in one or more cognitive domains that does not impair activities of daily living. The cognitive decline should not be due to delirium or other mental disorders.
DSM – 5-TR	Mild or Major Neurocognitive disorder	

Note. DSM-5-TR = Diagnostic Statistical Manual 5th edition [9], Text Revision; MMSE = Mini Mental State Examination; MDS = Movement Disorder Society; NCD = Neurocognitive Disorder; PD = Parkinson's disease; PDD = Parkinson's disease dementia; PD-MCI = Parkinson's disease Mild Cognitive Impairment.

for qualitative exploration of cognitive deficits [5]. Conversely, PD-MCI criteria allowed diagnosis using either level while acknowledging reduced diagnostic certainty for level 1 criteria [4]. By design, level 2 PD-MCI criteria therefore included clinical thresholds for cognitive measures to identify impairment, whereas level 2 PDD criteria did not. The threshold for impairment on a given test in the PD-MCI battery was defined as either: a) a score one to two standard deviations below demographically appropriate norms, b) a significant decline from previous performance on the test, or c) a significant decline from estimated premorbid levels.

Neuropsychological measures. The MDS-recommended level 2 test battery for PD-MCI comprised two tests for each of the five cognitive domains (memory, executive function, language, visuospatial processing, or attention/working memory). A diagnosis of PD-MCI is considered when impairment is demonstrated on two out of the 10 tests. Conversely, a diagnosis of PDD (or PD-MCI at level 1 criteria) following MDS criteria requires impairment in two cognitive domains, although this can be measured using a test of global cognitive function. Three other guidelines and one systematic review proposed alternative neuropsychological test batteries to detect cognitive disorders in PD [16,20,30,50]. Another systematic review reported tests sensitive to cognitive change over time in PD but fell short of recommending a neuropsychological test battery due to limited literature and the heterogeneity in tests used across studies [51]. Screening for behavioural and neuropsychiatric symptoms (e.g., depression, anxiety, and hallucinations) was also commonly recommended [4,5,16,20,21,27,44]. Several guidelines recommended specific assessments of functional ability [4,5,16,20,27,39], while others recommended informant interviews [4,16,39,44]. Recommended measures are provided in Table 3.

Frequency. Limited advice was given on how frequently cognitive testing for people living with PD should occur. Guidelines recommended that cognitive testing should occur regularly [21,37],

with baseline testing and regular re-assessment to track progression over time [24,37].

Motor considerations. Several guidelines emphasised that functional impairment due to cognitive impairment should be characterised carefully and be distinct from functional impairment due to the motor or autonomic symptoms of PD [4,5,16,22]. Three articles suggested that measures without heavy reliance on motor functions be used to assess cognition [4,5,16]. Three guidelines recommended tests be performed in the ON state, particularly for those with motor fluctuations [4,20,26].

Treatment

Pharmacological treatment. Table 4 summarises treatment recommendations for cognitive disorders in PD. Recommendations for the pharmacological treatment of PDD were consistent across articles. Cholinesterase inhibitors were most frequently recommended, with rivastigmine having the strongest overall evidence-base. There was mixed support for memantine, and weak or limited support for rasagiline and atomoxetine (Table 4). Four articles specifically addressed treatment for PD-MCI. One systematic review and two guidelines reported no efficacious pharmacological treatments for PD-MCI [23,33,52], while Connolly et al. (2014) reported inadequate evidence for atomoxetine to treat PD-MCI [29].

Non-pharmacological interventions. Although few guidelines provided recommendations for non-pharmacological interventions for cognitive disorders in PD, there was considerable evidence from systematic reviews. In particular, there was evidence of dance exercises improving cognitive in PD [53–55], as well as cognitive rehabilitation [56], dual task training [57], and yoga [58]. Exercise was strongly recommended in both guidelines and systematic reviews [18,21,33,54,59,60], with one systematic review further linking sedentary behaviour with increased risk of cognitive impairment in PD [61]. General lifestyle modifications such as cognitive activities,

Table 3

Recommended measures for neuropsychological evaluation in PD.

Citation	Purpose	Recommended tests
Neuropsychological test battery		
Dujardin et al. 2016	French consensus battery for identifying cognitive impairment in PD	Attention and working memory: Symbol Digit Modalities Test (oral), Digit Span Forward and Backward Episodic memory: 16-item Free/Cued Recall, 10/36 Spatial Recall Executive function: Letter-Number Sequencing (oral), D-KEFS Color-Word Interference Test Visuospatial function: Benton Judgement of Line Orientation, CLOX Language: Boston Naming Test, Animal fluency test (60 s)
Crizzle et al. 2012	Recommendations for a battery to identify cognitive impairment for driving assessment in PD	Level B evidence: The Useful Field of View test (total and subtest 2 scores), Trail-making (B and B-A), and Rey-Osterrieth Complex Figure test are probably predictive of driving performance. Level C evidence: Mini Mental State Examination (MMSE), Trail-making A, Clinical Dementia Rating scale, Hopkin's Verbal Learning Test, Judgement of Line Orientation, Wechsler Adult Intelligence Scale (WAIS), Benton Visual Retention Test, and Auditory Verbal Learning Test are possibly predictive of driving performance.
Severiano e Sousa et al. 2022	Recommended battery to assess cognition in Late-Stage PD	Attention and Executive Functions: Digit Span Backward, Trail-making (B-A), Semantic fluency Memory: Rey Auditory Verbal Learning Test, Brief Visual Memory Test -Revised Language: Semantic fluency, Boston Naming Test Visuospatial Functions: Cube copying, Brief Visual Memory Test -Revised, Judgement of Line Orientation, Clock-drawing Global: Montreal Cognitive Assessment (MoCA)
Jitkrisadakul et al. 2016	Recommendations for a battery to identify cognitive impairment for driving assessment in PD	Mini Mental State Examination (MMSE), Montreal Cognitive Assessment (MoCA), Trail A&B making test, Complex Figure Test, Block design test, Dot cancellation test
Litvan et al. 2012	MDS-recommended battery to identify MCI in PD (Level 2)	Attention and working memory: WAIS Letter Number Sequencing, WAIS Coding or other substitution task (written or oral), Trail-making Test, Digit span backward or digit ordering, STROOP color-word test Executive function: Wisconsin Card Sorting Test, or modified Card Sorting Test (Nelson's modification), Tower of London test–Drexel version, or Stockings of Cambridge (CANTAB), verbal fluency test (letter, category, or alternating fluency), Clock Drawing Test Language: WAIS Similarities, Confrontation naming (Boston Naming Test or Graded Naming Test) Memory: Word list learning test with delayed recall and recognition (Verbal learning tests or selective recall tests), prose recall test with a delayed recall (Wechsler Memory Scale-IV Logical Memory subtest or Rivermead Behavioural Memory Test), Brief Visuospatial Memory Test–Revised Visuospatial function: Benton Judgment of Line Orientation, Hooper Visual Organization Test, Clock copying (e.g., Royall's CLOX)
Bothe et al. 2010	Recommended battery to identify dementia in PD	Short-term/working memory: Digit span (forward, backward), Corsi block-tapping (forward, backward). Long-term memory: California Verbal Learning Test, Doors and people Test. Attention: TAP alertness, TAP go-no-go, TAP divided attention. Executive functions: Letter and category fluency, Raven's Colored Progressive Matrices (A and B), Clock-drawing. Visuospatial abilities: Visual Object and Space Perception (silhouettes and discriminate positions). Language: Boston Naming Test Executive functions: Digit span, Spatial span (CANTAB), Digit ordering, Similarities (WAIS-III), Wisconsin Card Sorting Test, Verbal fluency (C, F, L), Trail-making, STROOP, Odd man out test, Prehension behavior Memory: Auditory Verbal Learning Test, Free and cued recall test Language: Boston Naming Test, Copy of the clock Visuo-spatial: Benton Line Orientation Test, Visual Object and Space Perception (cube analysis). Visuo-perceptive: Benton Face Recognition Test, Visual Object and Space Perception (fragmented letters).
Dubois et al. 2007	MDS-recommended battery for monitoring dementia in PD (Level 2)	
Global cognitive tests		
Tripathi et al. 2022	Recommended tests to identify cognitive impairment in PD	Scales for Outcomes in Parkinson's disease (SCOPA-COG) Montreal Cognitive Assessment (MoCA) Mini Mental State Examination (MMSE)
Litvan et al. 2012	MDS-recommended tests to identify MCI in PD (level 1)	Montreal Cognitive Assessment (MoCA) Parkinson's disease Cognitive Rating Scale (PD-CRS) Scales for Outcomes in Parkinson's disease (SCOPA-COG) Mattis Dementia Ratings Scale (MDRS)
Parmera et al. 2022	Recommended tests to identify dementia in PD	Mini Mental State Examination (MMSE) preferably Montreal Cognitive Assessment (MoCA) or Addenbrooke's Cognitive Examination Revised (ACE-R)
Dubois et al. 2007	MDS-recommended tests to identify dementia in PD (level 1)	Mini Mental State Examination (MMSE) Mattis Dementia Ratings Scale (MDRS)
Instrumental Activities of Daily Living		
Dubois et al. 2007	Recommended tests to identify	Pill questionnaire or caregiver interview
Bothe et al. 2010	functional impairment due to	Pill questionnaire
Sorbi et al. 2012	cognition in PD	Caregiver interview
Dementia, NICE guideline [NG97] 2018		Informant Questionnaire on Cognitive Decline in the Elderly (IQCODE), Functional Activities Questionnaire (FAQ).
Parmera et al. 2022		Pfeffer Functional Activities Questionnaire (FAQ), the Disability Assessment for Dementia (DAD), the Informant Questionnaire on Cognitive Decline in the Elderly (IQCODE), Pill questionnaire, the phone call test, and the Direct Assessment of Functional Abilities (DAFA)

Note. D-KEFS = Delis Kaplan Executive Function System; WAIS = Wechsler Adult Intelligence Scale; CANTAB = Cambridge Neuropsychological Test Automated Battery.

Table 4

Recommendations for treatment of cognitive impairment in PD.

Treatment type	Treatment name	Guidelines	Systematic reviews
Acetyl cholinesterase inhibitors	Rivastigmine	Sufficient evidence[19,36]	Efficacious and clinically useful[52] Recommended (level 2) ^f [72]
		Effective (Class I) ^a [44]	
		Effective (Type 1) ^b [43]	
		Effective (Type 1 A) ^c [42]	
		Effective (Grade A) ^{a,d} [31,33]	
		Recommended[25,32]	
		Recommended (B) ^e [29]	
		Recommended (A) ^d [24]	
		Recommended at smaller doses than for Alzheimer's disease[18]	
		Should be used[23]	
	Donepezil	Sufficient evidence[19]	Recommended (level 1) ^f [72]
		Effective (Type 1) ^b [43]	
		Effective (Type 1 A) ^c [42]	
		Effective (Grade A) ^{a,d} [31,33]	
		Recommended[32]	
		Recommended (B) ^e [29]	
		Insufficient evidence[36]	
		Recommended off-label[25]	
		Recommended at smaller doses than for Alzheimer's disease[18]	
		Might be used[23]	
	Galantamine	Sufficient evidence[19]	Recommended (level 2) ^f [72]
		Insufficient evidence[36,44]	
		Effective (Type 1 A) ^c [42]	
		Effective (Grade C) ^{a,d} [31,33]	
		Recommended (U) ^e [29]	
		Recommended off-label[25]	
		Recommended at smaller doses than for Alzheimer's disease[18]	
		Cannot be used[23]	
		Insufficient evidence[19,36]	
		Probably effective (Level B) ^a [44]	
NMDA antagonists	Memantine	Possibly effective (Type 1) ^b [43]	No evidence for cognitive benefit (level 3) ^f [72]
		Possibly effective (Grade C) ^{a, d} [31,33]	
MAO-B inhibitors	Rasagiline	Recommended if Cholinesterase inhibitors are ineffective[25]	-
		No evidence for cognitive benefit[17]	
Selective norepinephrine reuptake inhibitors	Atomoxetine	Should not be used[23]	-
		Insufficient evidence[19,36]	
Non-pharmacological	Transcranial Direct Current Stimulation	Can improve cognition[18]	-
		Insufficient evidence[19]	
	Cognitive Rehabilitation	Recommended (U) ^e [29]	Probably effective[46]
		Insufficient evidence[19]	
	Cognitive Training (e.g., Computerised)	Probably effective (Level B) ^a [41]	Insufficient evidence[73]
		Insufficient evidence[19]	
	Dual Task training	Consider[39]	Moderate effect[56]
		Can be offered for PD-MCI[23]	
	Mental practice	Recommended[21]	Moderate effect[67]
		-	
	Dance exercise	Recommended[21]	Probable efficacy[74]
		-	
	Music contingent gait training	Recommended[21]	Moderate effect[57]
		-	
	Tai chi	Insufficient evidence[19]	Insufficient evidence[75]
		Probably effective (Level B) ^a [41]	
	Karate	Consider[39]	Possible benefits[53]
		Can be offered for PD-MCI[23]	
	Physical exercise	Recommended[21]	Recommended[54]
		-	
	Yoga	Encouraged for PD-MCI[33]	Recommended[55]
		Insufficient data for PDD[23]	
	Ketogenic diet	Should be conducted for PD-MCI[23]	Effective[60]
		-	
	Yoga	-	May be suitable[59]
		-	
	Ketogenic diet	-	Recommended[54]
		-	
	Ketogenic diet	-	Effective[60]
		-	

Note. Grading systems for quality of evidence and strength of recommendations:

^a EFNS guidelines

^b Shekelle et al. (1999) guidelines [77]

^c GRADE guidelines

^d SIGN guidelines

^e AAN guidelines

^f OCEBM Levels of Evidence Working Group

social engagement and diet (e.g., Mediterranean), were also recommended [18,21,24,33,39]. Finally, one guideline recommended intervention strategies such as cognitive stimulation to treat mild deterioration before establishing pharmacological treatment [24].

Special considerations. Special considerations were raised for the treatment of cognitive disorders in PD. Firstly, discontinuation of medications that influence cognition such as, anticholinergics [18,19,24,31,38], benzodiazepines [18,31], amantadine [18,24,31], selegiline [24], antipsychotics with high affinity for D2 receptors (e.g., haloperidol, risperidone) [19], tricyclic antidepressants [18,31,38], and tolterodine and oxybutynin (consider Mirabegron instead) [31,38]. NICE guidelines for dementia also highlighted that antipsychotics may worsen motor symptoms in patients with PDD and recommended that antipsychotics should be used at the lowest dose for the shortest possible time, reassessing their need every six weeks [39]. Secondly, Molina Ruiz et al. [18] acknowledged the potential ameliorative effects of dopaminergic treatment and atypical antipsychotics (quetiapine, clozapine) on cognition in PD, as well as the promise of nootropic medications such as piracetam [18]. Some considerations for the treatment of motor symptoms in people with PD and cognitive impairment were also recommended. Deep brain stimulation (DBS) was not recommended for those with mild cognitive problems [31,34], while levodopa-carbidopa intestinal gel infusions and subcutaneous apomorphine pumps were recommended and considered therapeutic and palliative for cognitively impaired people with PD ineligible for DBS [34].

Care

PD-specific care. Only one guideline provided specific care considerations for cognitive disorders in PD. Goldman et al. (2018) recommended that people with PD and cognitive disorders to take their time while completing tasks, let their support network know if they are having trouble, seek help if depressed or anxious, develop a highly structured daily routine to follow, have an advanced directive in place, and for care partners to ensure their own health is maintained [21]. For clinicians, they recommended referring people with PD and cognitive disorders to allied health early on. In particular, occupational therapists or psychologists were recommended for development of **cognitive** strategies (e.g., for attention), which was supported by other PD guidelines [24,37].

Driving. One guideline suggested that cognitive disorders in PD should provoke inquiry into driving safety with the potential of cessation [32], yet another recommended multidisciplinary evaluation, care partner input, and driving records to determine fitness to drive [30]. Three systematic reviews recommended cognitive assessments to evaluate driving ability, followed by mandatory retesting of driving skills for those with cognitive impairment [47,62], and one guideline recommended particular attention to be paid to visuospatial, visuo-perceptual, and executive abilities [44]. To aid physicians in determining driving ability, two systematic reviews provided explicit 'red flag' criteria for people with PD based on clinical history, sleepiness and impairments on motor, cognitive and visual assessments [62,63].

Dementia care. Most care recommendations were derived from general dementia guidelines. One guideline recommended clinicians refer to publicly available healthcare support programs for people living with cognitive disorders and arrange counselling and case/care management for care partners to reduce burden and increase satisfaction [44]. NICE guidelines recommended developing a care support plan, including discussions around advanced healthcare directives and enduring power of attorney, and that people living with dementia, family members and carers should have access dementia services, psychoeducation and skills training [39]. People with dementia should be informed about their care teams, contact details, local support services, workplace

adjustment rights, and any research studies that they can participate in. For people living with dementia experiencing agitation or aggression, personalised activities were recommended to promote engagement and pleasure.

Discussion

Summary of findings

The present review synthesised recommendations for the diagnosis, evaluation and management of cognitive disorders in PD. Diagnostic criteria and neuropsychological tests for the evaluation of cognitive disorders in PD were recommended by several articles, although recommendations from the MDS guidelines were cited most frequently. Treatment recommendations were relatively consistent, with endorsement of acetylcholinesterase inhibitors. However, there were some discrepancies in evidence for other pharmacotherapies and non-pharmacological interventions. Finally, there were limited care considerations, with most recommendations derived from guidelines for dementia.

Gaps in current literature

MDS guidelines from 2007 and 2012 were most highly cited among other guidelines and considered the current gold standard. However, given that these guidelines were published over a decade ago, further revision with updated literature is warranted. Furthermore, recommendations of cognitive measures for use in PD varied both within and between articles, with many cognitive tests recommended across various cognitive domains. A limitation of the current literature is therefore the lack of a harmonised toolkit for the evaluation of cognitive impairment in PD, which is corroborated by a meta-analysis that found high variability in cognitive measures used in PD and called for unified guidelines to improve comparability of test scores across cohorts [64].

Guidance on best practices for the process of neuropsychological evaluation for people with PD was limited. For example, it was unclear how frequently repeat neuropsychological assessment should be performed, how to proceed with subjective complaints without objective cognitive impairment, or whether to consider the assessment duration or test order. Some recommendations addressed the effect of motor fluctuations on cognitive performance, but questions remain about managing the impact of dyskinesia and distinguishing motor from cognitive symptoms in daily activities. None of the reviewed articles discussed practice effects and reliable change when conducting serial neuropsychological assessments, nor was there any consideration of relative cost to benefit of administering full neuropsychological assessments. Apart from MDS guidelines, there was also a lack of reference to cognitive subtyping. Cognitive subtypes may have clinical implications in PD, with impairment in certain domains predicting more rapid cognitive decline and eliciting unique care requirements (e.g., specialised cognitive training) [65,66].

There was also a distinct lack of recommendations for non-pharmacological interventions within the guidelines. While systematic reviews reported robust evidence for cognitive training and rehabilitation [56,67], they were endorsed by only three guidelines [23,37,39], suggesting a current gap between the literature and clinical practice. Similarly, care considerations were limited in the PD literature, lacking discussion of referral pathways or management of polypharmacy for people with PD and cognitive impairment. Finally, only four guidelines included input from stakeholders [21,25,26,39]. Involving peak professional bodies and lived experience experts ensures guidelines are practical and relevant for those with the target conditions [68]. The WHO endorses meaningful engagement of people with lived experience, which should be encouraged in future guideline development in PD [69].

Implications and future directions

The present investigation revealed several gaps in current guidelines for cognitive disorders in PD. Revised guidelines with up-to-date literature are called for, with novel attention given to processes of evaluating and diagnosing cognitive disorders as well as care considerations specific to PD. To improve guideline quality, greater involvement of stakeholders and consumers in the development of future guidelines is encouraged. Failure to evaluate and identify dementia in a timely manner presents safety issues to people living with PD and their families, and hinders early treatment, rehabilitation potential, support, and future planning. Given that there are few opportunities for cognitive evaluation in current PD clinical care, revised guidelines should aim to address the specific needs of people living with PD and cognitive disorders while encouraging adequate management and follow up procedures.

Limitations of present study

Limiting the search to English-language articles and publicly available grey literature may have restricted the scope of this review. Some systematic reviews had too few studies to offer clear recommendations due to insufficient evidence. Emphasis was placed on higher-quality articles, but the limitations of weaker reviews should be noted. Additionally, recommendations for DLB were excluded, following the "one-year rule" distinguishing DLB from PDD [70]. While DLB and PDD patients are often seen in different clinics, growing evidence suggests they may lie on a spectrum of alpha-synuclein diseases, rather than being separate entities [71].

Conclusions

The present review revealed important gaps and inconsistencies in current recommendations from guidelines and systematic reviews for the diagnosis, evaluation and management of cognitive disorders in PD. These gaps call for an up-to-date guideline with current evidence-based recommendations, providing specific advice for people living with PD experiencing, or at risk of, cognitive disorders.

Ethical compliance statement

The PDCogniCare project was approved by the Metro North Health HREC (HREC/2023/MNHA/100098). Informed patient consent was not necessary for this work.

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Financial interests, activities, relationships, and affiliations (past 12 months)

Dana Pourzinal is employed by The University of Queensland. Daniel Bailey does consultancy work for Department of Health (Residential Aged Care Star Ratings System), is employed by The University of Queensland, and has received grant funding from the NHMRC MRFF (PDCogniCare). Jihyun Yang is employed by the University of Queensland and has received grant funding from the NHMRC MRFF (PDCogniCare). Alexander Lehn provides expert testimony, is employed by, Queensland health, receives honoraria from SEER medical, Abbvie and Medtronic, and receives grant funding from Metro South Research Scheme. Helen Tinson is employed by Queensland Health. Jacki Liddle is employed by The University of

Queensland; Queensland Health, receives honoraria for PhD examination (Australian National University), and has received grant funding from the NHMRC, MRFF, Australian Association of Gerontology, and National Institute of Health Research. Deborah Brooks is employed by the University of Queensland and has received grant funding from the NHMRC MRFF (PDCogniCare) and Dementia Australia Research Foundation. Sharon Naismith does consultancy work for Eisai, Roche, is on the advisory board for Eisai, is employed by the University of Sydney, and receives grant funding from the NHMRC, Alzheimer's International, Alzheimer's Drug Discovery Foundation, and Medical Research Future Fund. Kirstine Shrubsole is employed by The University of Queensland and has received grant funding from the NHMRC Investigator Grant. Rodney Marsh is employed by the University of Queensland. Leander Mitchell is employed by The University of Queensland and has received grant funding from the NHMRC MRFF (PDCogniCare). Nancy Pachana is employed by The University of Queensland and has received grant funding from the NHMRC MRFF (PDCogniCare). James King is employed by the University of Queensland. Nadeeka Dissanayaka is employed by The University of Queensland and is honorary research fellow at the Royal Brisbane & Women's Hospital. She receives grant funding from NHMRC MRFF (PDCogniCare), NHMRC Boosting Dementia Research Leadership Fellowship, The Prince Charles Hospital Foundation Innovation Grant, and Dementia Australia Research Foundation Grants.

Author contributions

1. PDCogniCare Research project: A. Conception, B. Organization, C. Execution;
2. Systematic review: A. Search / screen, B. Data extraction, C. Quality assessment;
3. Manuscript Preparation: A. Writing of the first draft, B. Review and Critique.

DP: 1B, 1C, 2A, 2B, 2C, 3A; **CE:** 2A, 2B, 2C, 3A; **DXB:** 1C, 2A, 3B; **JY:** 1A, 3B; **AL:** 3B; **HT:** 3B; **JL:** 3B; **DB:** 3B; **SN:** 3B; **KS:** 3B; **RM:** 3B; **LKM:** 3B; **NAP:** 3B; **JK:** 3B; **NND:** 1A, 1B, 3B.

CRediT authorship contribution statement

Pourzinal Dana: Data curation, Formal analysis, Investigation, Methodology, Project administration, Supervision, Visualization, Writing – original draft, Writing – review & editing. **Elgey Charlotte:** Conceptualization, Writing – review & editing. **Bailey Daniel X.:** Formal analysis, Writing – review & editing. **Yang Jihyun:** Formal analysis, Writing – review & editing. **Lehn Alexander:** Writing – review & editing. **Tinson Helen:** Writing – review & editing. **Liddle Jacki:** Writing – review & editing. **Brooks Deborah:** Conceptualization, Writing – review & editing, Project administration, Supervision. **Naismith Sharon L.:** Conceptualization, Writing – review & editing. **Shrubsole Kirstine:** Writing – review & editing. **Marsh Rodney:** Writing – review & editing. **Mitchell Leander K.:** Conceptualization, Writing – review & editing. **Pachana Nancy A.:** Writing – review & editing. **King James:** Formal analysis, Writing – review & editing. **Dissanayaka N. Nadeeka:** Conceptualization, Project administration, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Nadeeka Dissanayaka reports financial support was provided by National Health and Medical Research Council. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data sharing statement

No original data was produced for this manuscript. Data extraction files may be shared, upon request.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.inpsyc.2025.100081](https://doi.org/10.1016/j.inpsyc.2025.100081).

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